

Ian Vandevent

ivandevent@gmail.com · +1 (805) 712-9887 · ianvandevent.com · github.com/ivandevent

Summary

Data scientist and software engineer with a Ph.D. in seismology (UC San Diego, 2026), specializing in analysis pipelines for datasets exceeding 10 million records. Real-world and academic experience spans the full lifecycle of scientific computing: ingesting and cleaning messy real-world data, designing efficient numerical algorithms, and building analysis and visualization workflows. Author of peer-reviewed publications and open-source scientific tools.

Education

Ph.D., Geophysics — Scripps Institution of Oceanography, UC San Diego	2021–2026
<i>Earthquake source properties — Profs. Peter Shearer, Wenyuan Fan</i>	<i>San Diego, CA</i>
B.S., Earth Science (Geophysics emphasis) — UC Santa Barbara	2016–2019

Technical Skills

Languages & Tools: Python (NumPy, (geo)pandas, SciPy, ObsPy), GMT, QGIS, MATLAB, Fortran
Methods: Data inversion, spectral analysis, signal processing, time-series analysis, spatial statistics, concurrent/parallel processing
Other: Linux/macOS/Windows, Office suite, git version control, remote workflows

Experience

Graduate Student Researcher	2021 – 2026
<i>Scripps Institution of Oceanography, UC San Diego</i>	<i>San Diego, CA</i>

- Developed methods for characterizing earthquake source properties, applied to 30 years of California seismicity (~775,000 events, ~12 million seismograms)
- Introduced the normalized frequency index (nFI), a metric distinguishing spatial variations in earthquake source physics; released as an open-source end-to-end Python pipeline (`py-nfi`, below)
- Contributed to the SCEC/USGS Community Stress Drop Validation Study, a multi-institution comparison of stress-drop estimation methods for the 2019 Ridgecrest sequence
- 4 peer-reviewed publications (plus one submitted) and 12 conference presentations (AGU, SCEC, SSA, EGU); two outstanding presentation awards (AGU, 2024; EGU, 2026)

Geotechnical Associate	2019 – 2020
<i>C2Earth, Inc.</i>	<i>Campbell, CA</i>

- Built an internal project-tracking web application (HTML/JS frontend, SQL backend); replaced a single overloaded spreadsheet of thousands of projects with searchable, editable records adopted firm-wide
- Supported construction projects through field and laboratory soil testing and settlement contour mapping

Software

py-nfi (github.com/ivandevent/py-nfi)	2025 – present
<i>Computes the normalized frequency index for large seismic catalogs</i>	<i>Python</i>

- Processes earthquake spectra: source effect isolation, magnitude correction, and uncertainty estimation. Multiprocessing enabled for efficient processing of large datasets: 12M records processed in ~ 8 minutes.

seisplot (github.com/ivandevent/seisplot)	2022
<i>Interactive GUI for seismogram visualization, filtering, and quality control, with a plugin architecture.</i>	<i>Python</i>

Selected Publications

1. **Vandevent, I.**, Shearer, P.M., Fan, W. “Using a High- to Low-Frequency Spectral Ratio to Distinguish Variations in Earthquake Source Properties.” *BSSA*, 2026. [doi.org/10.1785/0120250171]
2. **Vandevent, I.**, Shearer, P.M., Fan, W. “Ridgecrest Aftershock Stress Drops from P- and S-Wave Spectral Decomposition.” *BSSA*, 2024. [doi.org/10.1785/0120240133]
3. Co-author on two additional *BSSA* publications; full list at ianvandevent.com.